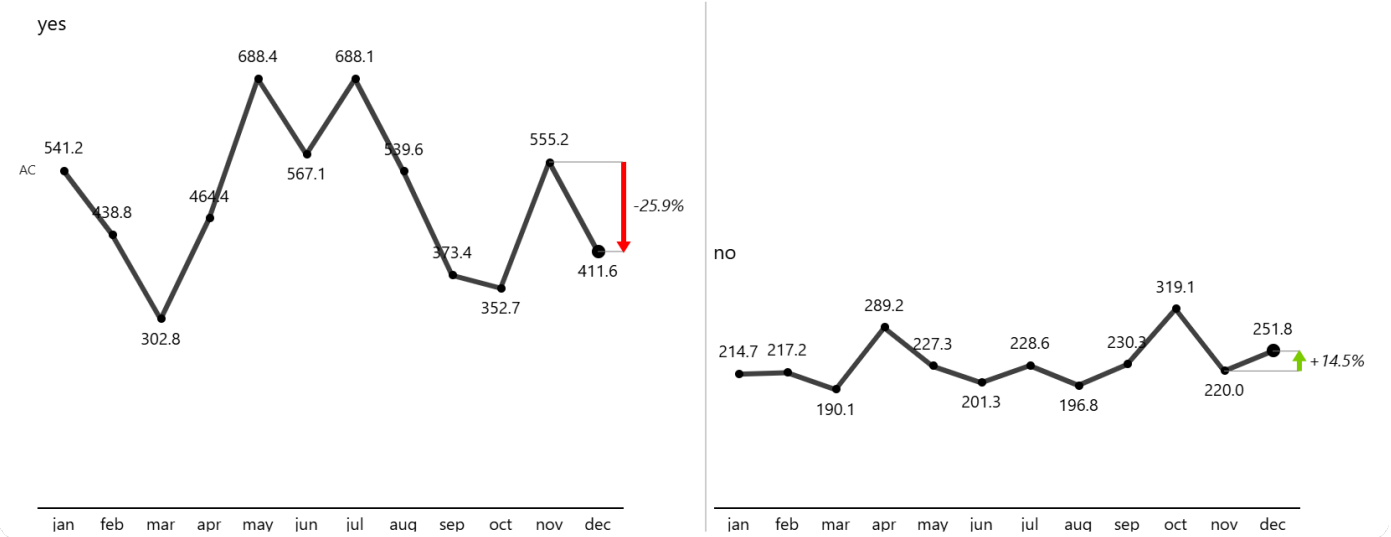
Longer calls and fewer contact attempts mark deposit subscribers, with previous successes, retirees and students converting best

MTD

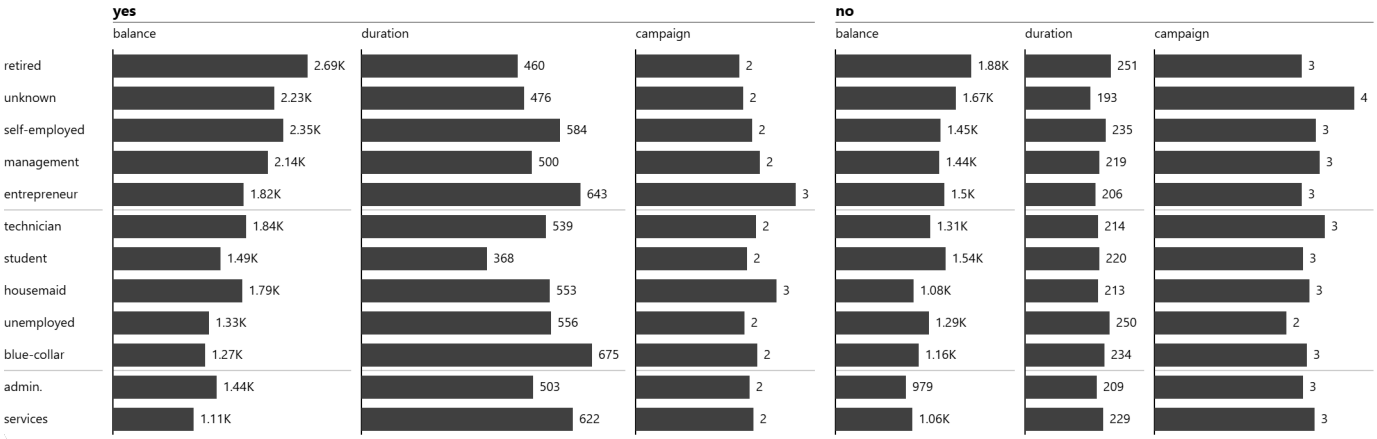
Subscribers talk about 2.4x longer than non-subscribers, averaging 688 seconds in May and July versus about 200–230 for non-subscribers

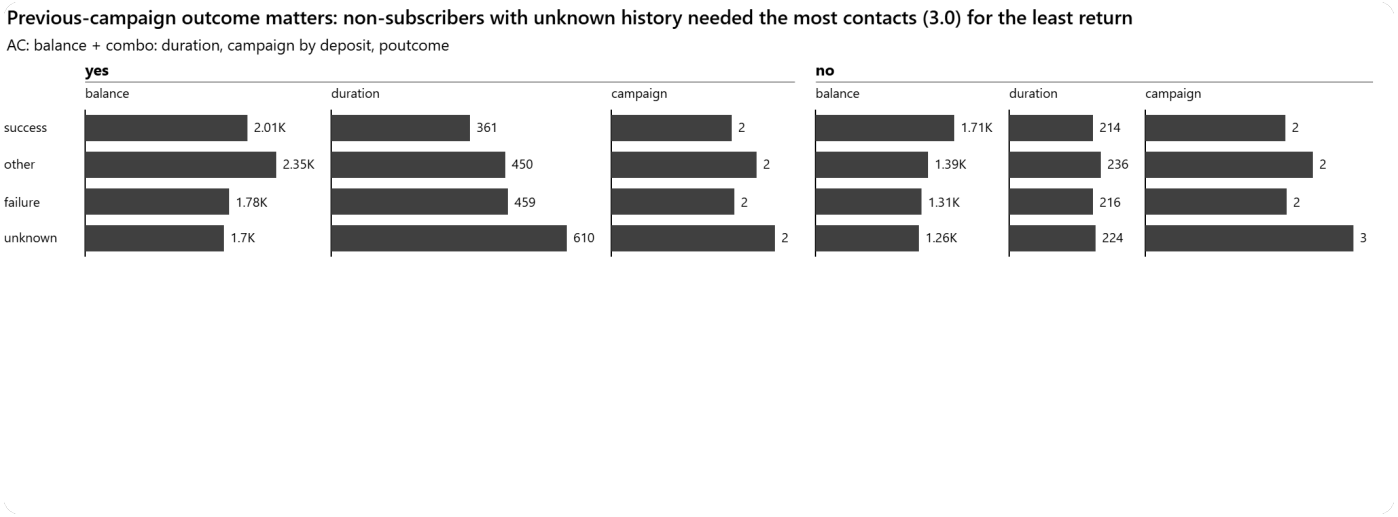
AC: duration by deposit, month



Retired subscribers hold the highest average balance (2.7K) and need the fewest contacts (1.9), while non-subscribers need about 2.4–3.1

AC: balance + combo: duration, campaign by deposit, job





Summary

Call engagement separates subscribers from non-subscribers. Subscribers average about 537s per call, against about 223s for non-subscribers. They also need fewer contacts (2.1 vs 2.8) and hold higher balances (1.8K vs 1.3K). Subscription rates are lowest in May (33%) and highest in Mar, Sep, Oct and Dec (82–91%). Rates are 91% after a previous success, 66% for retired customers, and 75% for students. Rates are only 23% where the contact channel is unknown. Customers with a housing loan subscribe less often (37% vs 57%).

Advice

- Prioritise customers with a previous campaign success, plus retired and student segments.
- Cap repeat contacts for customers who are not engaging. Non-subscribers average about 2.8 attempts for little return.
- Move more effort out of May into higher-converting months, and test whether the low rates there come from volume.
- Record the contact channel. Unknown-channel contacts convert poorly.
- Treat call duration as a signal of engagement, not a cause. It is only known after the call.

Analyze the Bank Marketing dataset.

1. Find total customers.
2. Identify subscribed customers.
3. Find customers who did not subscribe.
4. Analyze customer age and job.
5. Find the best marketing contact method.
6. Analyze loan and marital status.
7. Calculate subscription rate.
8. Create simple charts and visualizations.
9. Show important business insights.
10. Give suggestions to improve marketing results.

Summary

Headline numbers

Question	Result
Total customers	11,162

Question	Result
Subscribed	5,289
Did not subscribe	5,873
Overall subscription rate	47.4%

Age and job

- Average age is nearly identical for subscribers (41.7) and non-subscribers (40.8). Age matters as a segment, not as an average.
- The 25-34 and 35-44 groups make up 63% of customers, but they convert at only 49% and 42%.
- The 65+ group converts at 80.2% and the under-25 group at 73.4%. Both are small (425 and 282 customers).
- Students (74.7%) and retired customers (66.3%) convert best. Blue-collar (36.4%) and entrepreneurs (37.5%) convert worst.

Best contact method

- Cellular is best. It reaches 8,042 customers at a 54.3% rate and produces 83% of all subscriptions.
- Telephone converts at 50.4%, but only 774 customers were contacted this way.
- Unknown channel converts at only 22.6%.

Loan and marital status

- Customers with no loan subscribe at 49.5%, against 33.2% for those with a loan.
- Single customers convert best (54.3%), then divorced (48.1%) and married (43.4%). Married customers are the largest group (6,351).

Suggestions

- Prioritise cellular outreach and students, retired customers and the 65+ and under-25 groups.
- Make sure the contact channel is recorded. Unknown-channel contacts convert poorly, so fix the data capture and the channel mix.
- Use fewer resources on customers with existing loans, and on blue-collar and entrepreneur segments. Test a different offer for them instead.
- Create a tailored offer for married customers. They are the largest group and the weakest converters, so small gains there add the most volume.
- Check whether the 25-44 segment needs a different message, since it holds the most customers but converts below average.

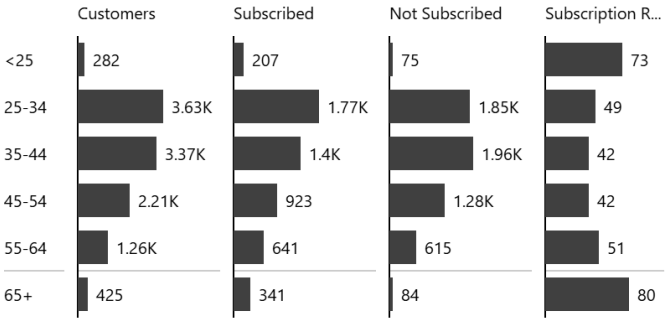
Method and notes

I counted each row as one customer and treated deposit = yes as subscribed. The subscription rate is subscribed customers divided by total customers for each segment. I grouped age into bands (<25, 25-34, 35-44, 45-54, 55-64, 65+). The dataset has no year column, so no time filter applied. No rows were removed. unknown values in contact and job were kept as their own categories.

Would you like me to look at how the month or previous campaign outcome (poutcome) combine with contact method to refine these suggestions?

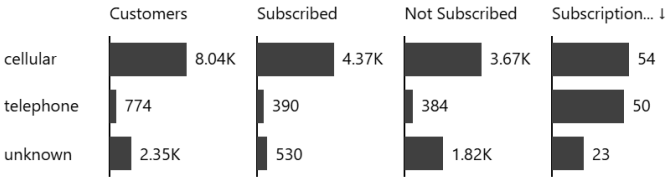
25-34 and 35-44 make up 63% of customers, but 65+ has the highest subscription rate at 80.2% and <25 follows at 73.4%

AC: Customers + Subscribed + Not Subscribed + Subscription Rate %
by age_group



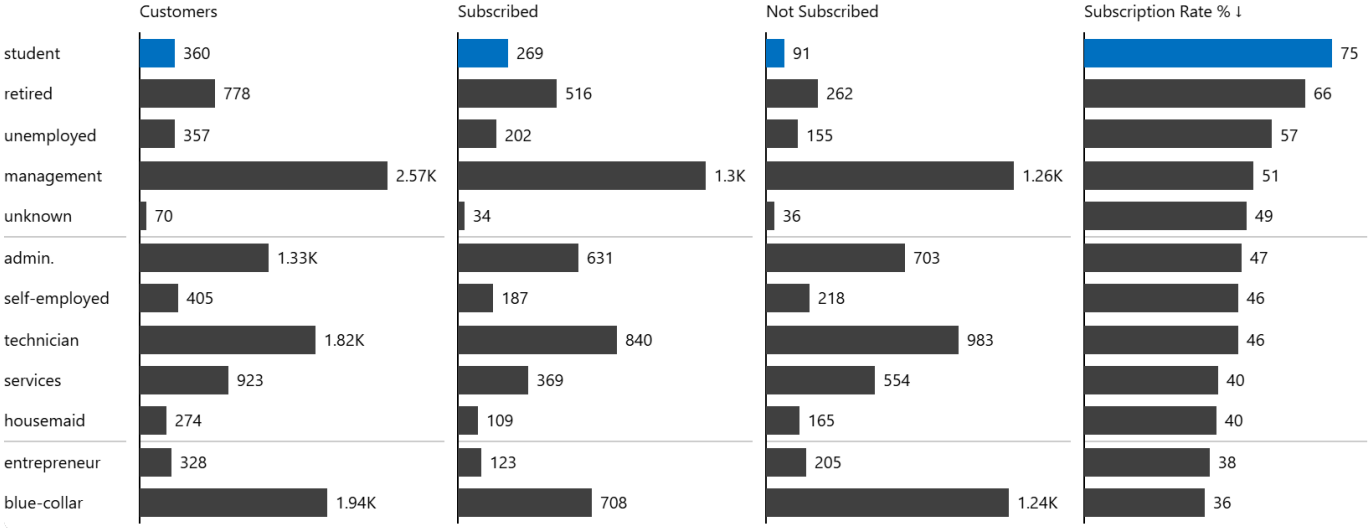
Cellular drives 8.0K customers and 4.4K subscriptions, while unknown contact lags with just 22.6% subscription rate

AC: Customers + Subscribed + Not Subscribed + Subscription Rate %
by contact



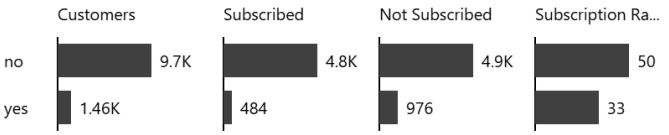
Students have the highest subscription rate at 74.7%, while blue-collar and management account for the most non-subscribed customers.

AC: Customers + Subscribed + Not Subscribed + Subscription Rate % by job



No-loan customers dominate at 9.7K, with a higher subscription rate than loan customers (49.5% vs 33.2%)

AC: Customers + Subscribed + Not Subscribed + Subscription Rate %
by loan



Married customers are the largest group (6.4K) but have the lowest subscription rate at 43.4%, while singles lead at 54.3%

AC: Customers + Subscribed + Not Subscribed + Subscription Rate %
by marital

